

EXPERIMENT-6 16-BIT SUBTRACTION

Aim: To write the assembly language program to implement 16-bit subtraction using 8085 processor.

Algorithm;

1. Start the program by loading a register pair with address of 1st number.
2. Copy the data to another register pair.
3. Load the second number to the first register pair.
4. Sub the two register pair contents.
5. Check for ~~for~~ carry.
6. Store the value of sum and carry in memory locations.
7. End.

Program:

1. LHLD 2050 - load the 16-bit data from memory locations 2050H & 2051H into the HL register pair.
2. XCHG - Exchange the contents of the HL & DE register pairs.
3. LHLD 2052 - load the 16-bit data from memory locations 2052H and 2053H into the HL register pair.
4. MVI ~~C~~, 00 - Move the immediate value 00H into register C.
5. MOV A, E - Move the immediate value 00H into the accumulator (A).
6. SUB L - subtract the content of register L from the accumulator (A).
7. STA 2054 - store the result of the subtraction into memory location 2054H.

8. MOV A, D - move the content of register D into the accumulator (A).
9. SUB H - subtract the content of register H from the accumulator (A).
10. STA 2055 - store the result of the subtraction into memory location 2055H.
11. HLT - ~~Halt~~ the execution of the program.

Input:

Address	Data
2050	2
2052	3

Output:

Address	Data
2054	1
2055	1

Result: Thus the program was executed successfully using 8085 processor simulator.

FileResetAssemblerDebugHelp

Registers

A	00
BC	05 00
DE	00 03
HL	00 02
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

Flag

S	0
Z	1
AC	0
P	1
C	0

Load me at:

1

LHLD

2050

2

XCHG

3

LHLD

2052

4

MVI

C,00

5

MOV

A,E

6

SUB

L

7

STA

2054

8

MOV

A,D

9

SUB

H

10

STA

2055

11

HLT

Decimal - Hex Conversion

Decimal

Hex

0

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

2052

-

+

02

Update Memory

GNUSim8085 - 8085 Microprocessor Simulator

DataStackKeyPadMemoryI/O Ports

Start2054OK

Address (Hex)	Address	Data
0806	2054	1
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0
0810	2064	0
0811	2065	0
0812	2066	0
0813	2067	0
0814	2068	0
0815	2069	0

Line NoAssembler Message

0Program assembled successfully